

Generation Interconnection System Impact Study Report

For Queue Project Q16-0584

West Waynesboro 138 kV

May 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0584. The Interconnection Customer proposes a 154.0 MW battery energy storage facility interconnecting at the West Waynesboro 138 kV substation in the Pennsylvania area. The request was received on February 25, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 277 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,654,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hutchinson Plant 1 – Yacht Club D.S 138kV	Watford City – Hutchinson Plant 1 138kV	142.0	282	5.1
Yacht Club D.S – Orrtanna 138kV	Hutchinson Plant 1 – Watford City 138kV	134.9	292	5.1
Altavista Power Sta – Catoctin 138kV	W Kelso – Dry Gap 138kV	132.1	260	6.4
Antrim – Peterson Flats 138kV	Chambersburg – W Waynesboro 345kV L	129.8	264	12.4
Otsego – Peterson Flats 138kV	W Waynesboro – Chambersburg 345kV	129.1	253	12.4
Thurmont – Catoctin 138kV	West Kelso – Dry Gap 138kV	128.2	285	6.4
Chambersburg 345/138 kV Transformer	Chambersburg – West Waynesboro 345kV Line	126.8	503	18.0
Ess Wesel – Pineville 138kV line	Chambersburg – W Waynesboro 345kV line	125.6	277	24.1
M105 – Wyoming 138kV	Chambersburg Substation – West Waynesboro 345kV	118.5	394	19.9
Wyoming – N Lewiston 138kV	W Waynesboro – Chambersburg 345kV	110.1	441	19.9
Carrel – Buffalo Tungsten 138kV line	Fair Oaks – Chambersburg 345kV L	109.1	449	5.3
Cressey Substation – Ess Wesel 138kV	Chambersburg – West Waynesboro 345kV	109.1	351	24.1
Chevmain – Cressey Substation 138kV L	Chambersburg – Fair Oaks 345kV Line	106.4	447	5.3
N Lewiston – Chambersburg 138kV	Chambersburg – West Waynesboro 345kV	104.8	492	13.9
Four Hills – Rubidoux 138kV	West Kelso – Dry Gap 138kV line	103.8	540	6.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Altavista Power Station – Rubidoux 138kV	Dry Gap – W Kelso 138kV	103.0	568	6.8
Carrel – Lourich 138kV	Fair Oaks – Chambersburg 345kV	102.2	376	9.3
Thurmont – Dry Gap 138kV line	West Kelso Substation – Dry Gap 138kV	102.1	459	11.4
Buffalo Tungsten – Chevmain 138kV	Fair Oks – Chambersburg 345kV	100.8	479	5.3
Hutchinson Plant 1 – Watford City 138kV	Yacht Club D.S – Hutchinson Plant 1 138kV Line	100.3	604	7.7
Antrim – Mtn View 138kV	Fair Oaks – Chambersburg Substation 345kV	100.2	434	9.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yacht Club D.S – Hutchinson Plant 1 138kV	Rebuild line with higher-capacity conductor	\$14,350,000	36% (\$5,175,000)
Yacht Club D.S – Orrtanna 138kV	Rebuild line with higher-capacity conductor	\$3,079,000	36% (\$1,110,000)
Altavista Power Station – Catoctin 138kV L	Rebuild line with higher-capacity conductor	\$4,086,000	40% (\$1,619,000)
Antrim – Peterson Flats 138kV	Rebuild line with higher-capacity conductor	\$4,926,000	35% (\$1,730,000)
Otsego – Peterson Flats 138kV	Rebuild line with higher-capacity conductor	\$3,015,000	35% (\$1,059,000)
Catoctin – Thurmont 138kV	Rebuild line with higher-capacity conductor	\$5,992,000	40% (\$2,374,000)
Chambersburg 345/138 kV Transformer	Replace transformer with higher-rated unit	\$19,756,000	11% (\$2,100,000)
Ess Wesel – Pineville 138kV	Rebuild line with higher-capacity conductor	\$4,518,000	43% (\$1,941,000)
M105 – Wyoming 138kV	Rebuild line with higher-capacity conductor	\$1,135,000	16% (\$184,000)
Wyoming – N Lewiston 138kV	Rebuild line with higher-capacity conductor	\$972,000	16% (\$157,000)
Carrel – Buffalo Tungsten 138kV	Reconductor line	\$616,000	4% (\$22,000)
Cressey Substation – Ess Wesel 138kV	Reconductor line	\$2,012,000	43% (\$865,000)
Chevmain – Cressey Substation 138kV L	Reconductor line	\$3,102,000	4% (\$110,000)
N Lewiston – Chambersburg 138kV	Reconductor line	\$2,053,000	10% (\$210,000)
Four Hills – Rubidoux 138kV L	Reconductor line	\$3,098,000	29% (\$890,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Altavista Power Sta – Rubidoux 138kV	Reconductor line	\$755,000	29% (\$217,000)
Carrel – Lourich 138kV	Reconductor line	\$3,706,000	6% (\$214,000)
Thurmont – Dry Gap 138kV line	Reconductor line	\$2,045,000	11% (\$222,000)
Buffalo Tungsten – Chevmain 138kV	Reconductor line	\$1,231,000	4% (\$43,000)
Hutchinson Plant 1 – Watford City 138kV	Reconductor line	\$740,000	32% (\$238,000)
Antrim – Mtn View 138kV	Reconductor line	\$2,849,000	6% (\$174,000)
	Total allocated to project		\$20,654,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.